

1.

 **SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C++ PROGRAMMING

QUESTIONS 19 / 19 completed

Sort

SORT

Write a C++ program to overload a function to sort an integer array and a double array separately.

PROGRAM EDITOR

C++

Run Save

```
1 #include <iostream>
2 #include <algorithm>
3 using namespace std;
4
5 // Overloaded function to sort an integer array
6 void sortArray(int arr[], int size) {
7     sort(arr, arr + size);
8     cout << "Sorted Integer Array: ";
9     for (int i = 0; i < size; i++) {
10         cout << arr[i] << " ";
11     }
12     cout << endl;
13 }
14
15 // Overloaded function to sort a double array
16 void sortArray(double arr[], int size) {
17     sort(arr, arr + size);
18     cout << "Sorted Double Array: ";
19     for (int i = 0; i < size; i++) {
20         cout << arr[i] << " ";
21     }
```

2.

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the course "C++ PROGRAMMING", and a user profile for "SHRUTHI SREE R" with ID "192421129RC". The left sidebar shows "QUESTIONS 19 / 19 completed" and a dropdown menu with "Power" selected. The main area is the "PROGRAM EDITOR" with a C++ code editor. The code implements two overloaded functions: one for integer base and exponent, and another for double base and exponent. The right sidebar shows the "INPUT" field with the values "2 5" and "2.5 3.0".

QUESTIONS 19 / 19 completed

Power

POWER

Write a C++ program to overload a function to calculate the power of an integer number and the power of a floating-point number separately.

PROGRAM EDITOR C++ Run Save

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5 // Overloaded function to calculate power for integer base and exponent
6 long long power(int base, int exp) {
7     long long result = 1;
8     for (int i = 0; i < exp; i++) {
9         result *= base;
10    }
11    return result;
12 }
13
14 // Overloaded function to calculate power for double base and exponent
15 double power(double base, double exp) {
16     return pow(base, exp);
17 }
18
19 int main() {
20     int baseInt, expInt;
21     double baseDouble, expDouble;
```

INPUT

2 5
2.5 3.0

3.

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", the course "C++ PROGRAMMING", and a user profile for "SHRUTHI SREE R" with ID "192421129RC". The left sidebar shows "QUESTIONS 19 / 19 completed" and a dropdown menu with "Absolute" selected. The main area is the "PROGRAM EDITOR" with a C++ code editor. The code implements two overloaded functions: one for integer and another for double. The right sidebar shows the "INPUT" field with the values "-15" and "-8.75".

QUESTIONS 19 / 19 completed

Absolute

ABSOLUTE

Write a C++ program to overload a function to find the absolute value of an integer number and the absolute value of a floating-point number separately.

PROGRAM EDITOR C++ Run Save

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5 // Overloaded function to find absolute value of an integer
6 int absolute(int n) {
7     return abs(n);
8 }
9
10 // Overloaded function to find absolute value of a floating-point number
11 double absolute(double n) {
12     return fabs(n);
13 }
14
15 int main() {
16     int i_num;
17     double f_num;
18
19     cin >> i_num;
20     cin >> f_num;
21 }
```

INPUT

-15
-8.75

4.



SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS 9 / 9 completed

Factorial-Fun. ▾

FACTORIAL-FUN.

Write a function to compute the factorial of a given integer.

PROGRAM EDITOR C++ ▾ Run Save

```
1 #include <iostream>
2 using namespace std;
3
4 // Function to compute factorial of a given integer
5 long long factorial(int n) {
6     long long fact = 1;
7     for (int i = 1; i <= n; i++) {
8         fact *= i;
9     }
10    return fact;
11 }
12
13 int main() {
14     int n;
15     cin >> n;
16
17     cout << factorial(n) << endl;
18
19     return 0;
20 }
```

INPUT

5

5.



SIMATS Online Programming Lab
C++ PROGRAMMING

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QUESTIONS 9 / 9 completed

Prime-Fun. ▾

PRIME-FUN.

Write a function to determine if a given integer is a prime number or not.

PROGRAM EDITOR C++ ▾ Run Save

```
1 #include <iostream>
2 using namespace std;
3
4 // Function to check if a number is prime
5 bool isPrime(int n) {
6     if (n <= 1) return false;
7     for (int i = 2; i * i <= n; i++) {
8         if (n % i == 0) return false;
9     }
10    return true;
11 }
12
13 int main() {
14     int num;
15     cin >> num;
16
17     if (isPrime(num)) {
18         cout << num << " is a prime number." << endl;
19     } else {
20         cout << num << " is not a prime number." << endl;
21     }
22 }
```

INPUT

7